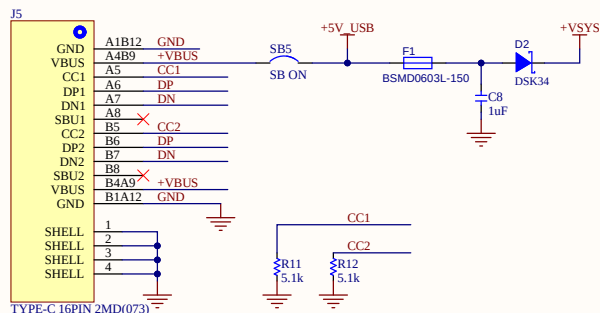


1 POWER

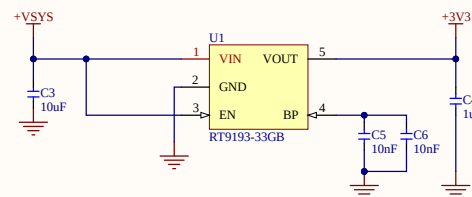
Power.SchDoc

MCU
PA9/TX1 >> PA9/TX1
PA10/RX1 >> PA10/RX1
CH_RX >> CH_RX
CH_TX >> CH_TX

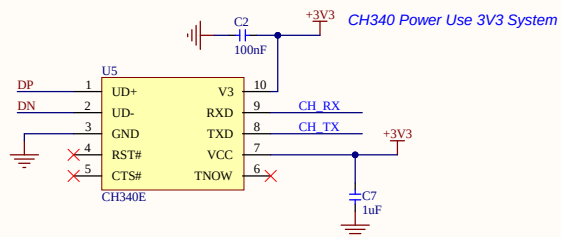
USB Type C



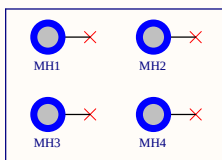
5v to 3v3@100mA



CH340E USB to TTL



3.2mm mounting hole without copper



ESD

Logo STM32

Logo Grove Module

Logo EPCB

AK BASE KIT 2.2

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2 RS485

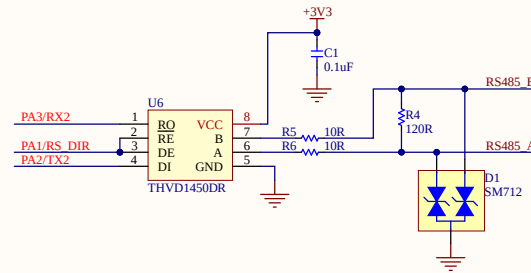
RS485.SchDoc

RS485

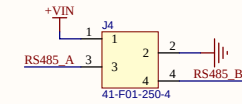
PA3/RX2 >> PA3/RX2
PA1/RS_DIR >> PA1/RS_DIR
PA2/TX2 >> PA2/TX2

PA3/RX2
T1
PA1/RS_DIR
T2
PA2/TX2
T3

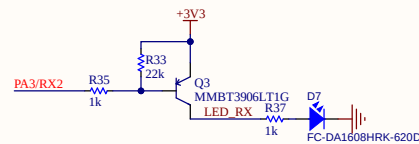
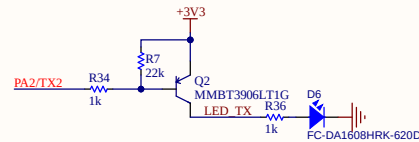
RS485 IC



Spring Clamp Connector



RS485 RX,TX LED Indicator



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3 NRF24L01

NRF24L01.SchDoc

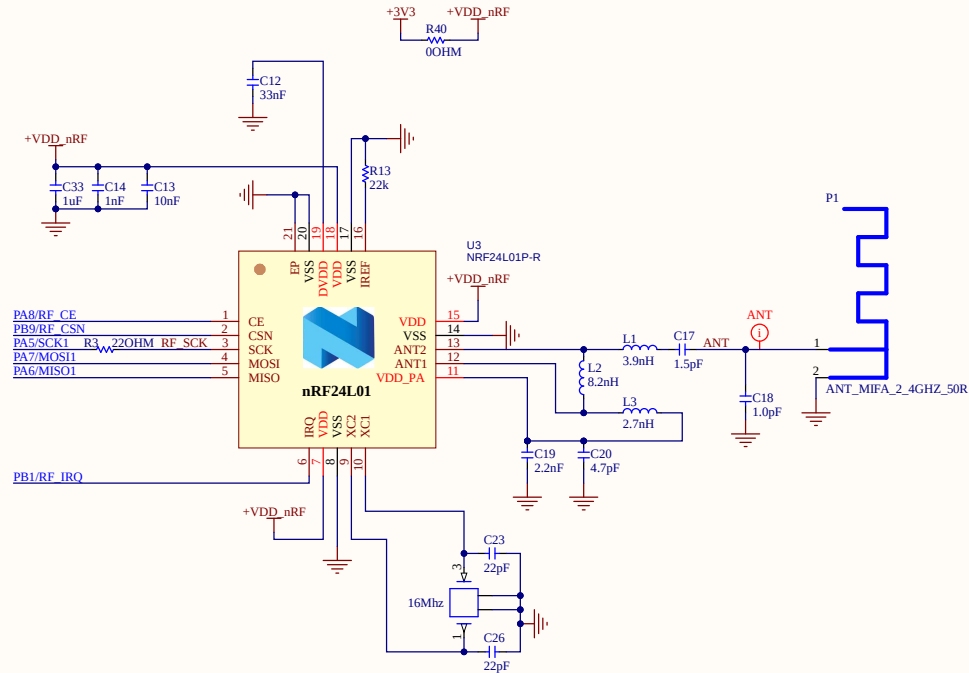
NRF24L01

PA8/RF_CE	>>	PA8/RF_CE
PB9/RF_CSN	>>	PB9/RF_CSN
PA5/SCK1	>>	PA5/SCK1
PA7/MOSI1	>>	PA7/MOSI1
PA6/MISO1	>>	PA6/MISO1
PB1/RF_IRQ	>>	PB1/RF_IRQ

Test Point nRF

T4	PA8/RF_CE
T5	PB9/RF_CSN
T6	RF_SCK
T7	PA7/MOSI1
T8	PA6/MISO1
T9	PB1/RF_IRQ

NRF24L01



AK BASE KIT 2.2

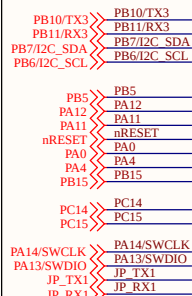
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Page Contents:

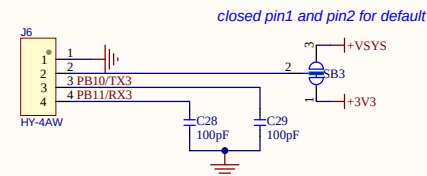
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Interface.SchDoc

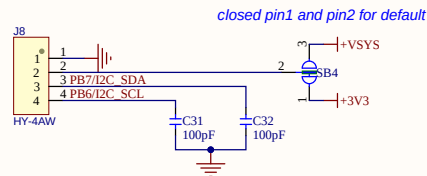
MCU



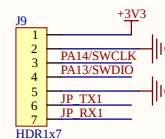
GROVE UART



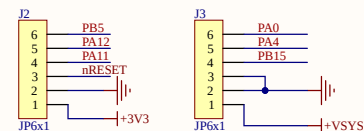
GROVE I2C



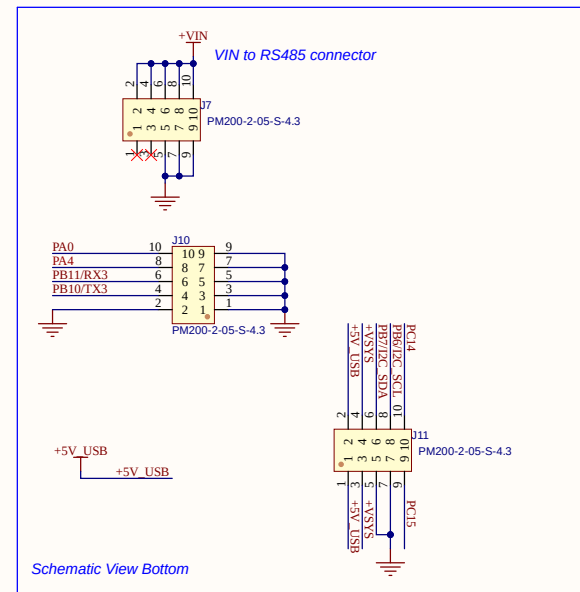
DEBUG



Header Male 2.54mm pinout (Default header not mount)



Expansion Board Connector



Schematic View Bottom

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5 MCU

MCU.SchDoc

NRF24L01

PA8/RX_CE >> PA8/RX_CE
PB9/RX_CSN >> PB9/RX_CSN
PA5/SCK1 >> PA5/SCK1
PA7/MOSI1 >> PA7/MOSI1
PA6/MISO1 >> PA6/MISO1
PB1/RX_IRQ >> PB1/RX_IRQ

RS485

PA3/RX2 >> PA3/RX2
PA1/RX_DIR >> PA1/RX_DIR
PA2/TX2 >> PA2/TX2

Power

PA9/TX1 >> PA9/TX1
PA10/RX1 >> PA10/RX1
CH_RX >> CH_RX
CH_TX >> CH_TX

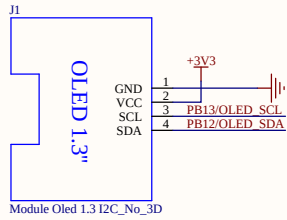
Interface

PB10/TX3 >> PB10/TX3
PB11/RX3 >> PB11/RX3
PB7/I2C_SDA >> PB7/I2C_SDA
PB6/I2C_SCL >> PB6/I2C_SCL

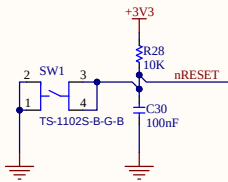
PB5 >> PB5
PA12 >> PA12
PA11 >> PA11
nRESET >> nRESET
PA0 >> PA0
PA4 >> PA4
PB15 >> PB15

PA14/SWCLK >> PA14/SWCLK
PA13/SWDIO >> PA13/SWDIO
JP_TX1 >> JP_TX1
JP_RX1 >> JP_RX1
PC14 >> PC14
PC15 >> PC15

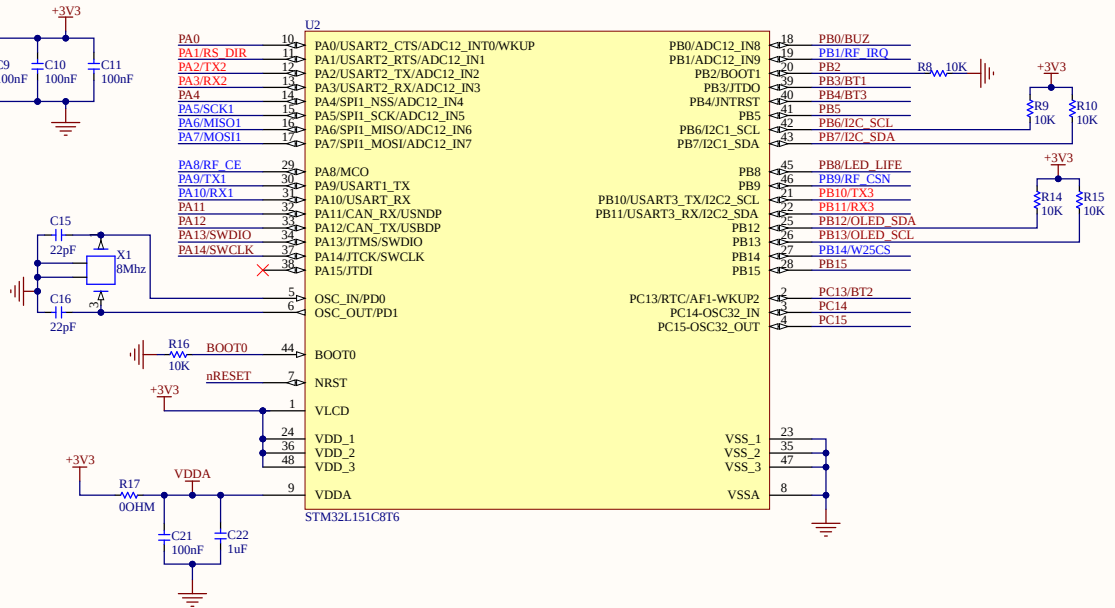
Module LCD Oled I2C



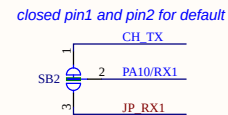
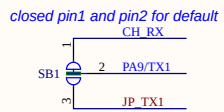
Reset



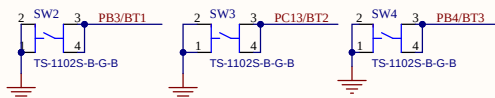
STM32L151C8T6



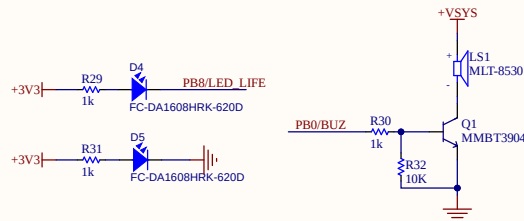
Solder Bridge UART



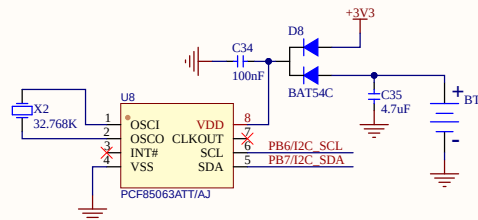
Button



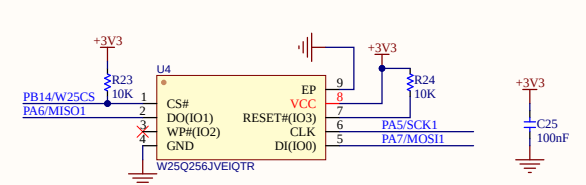
Led & Buzzer



RTC PCF85063

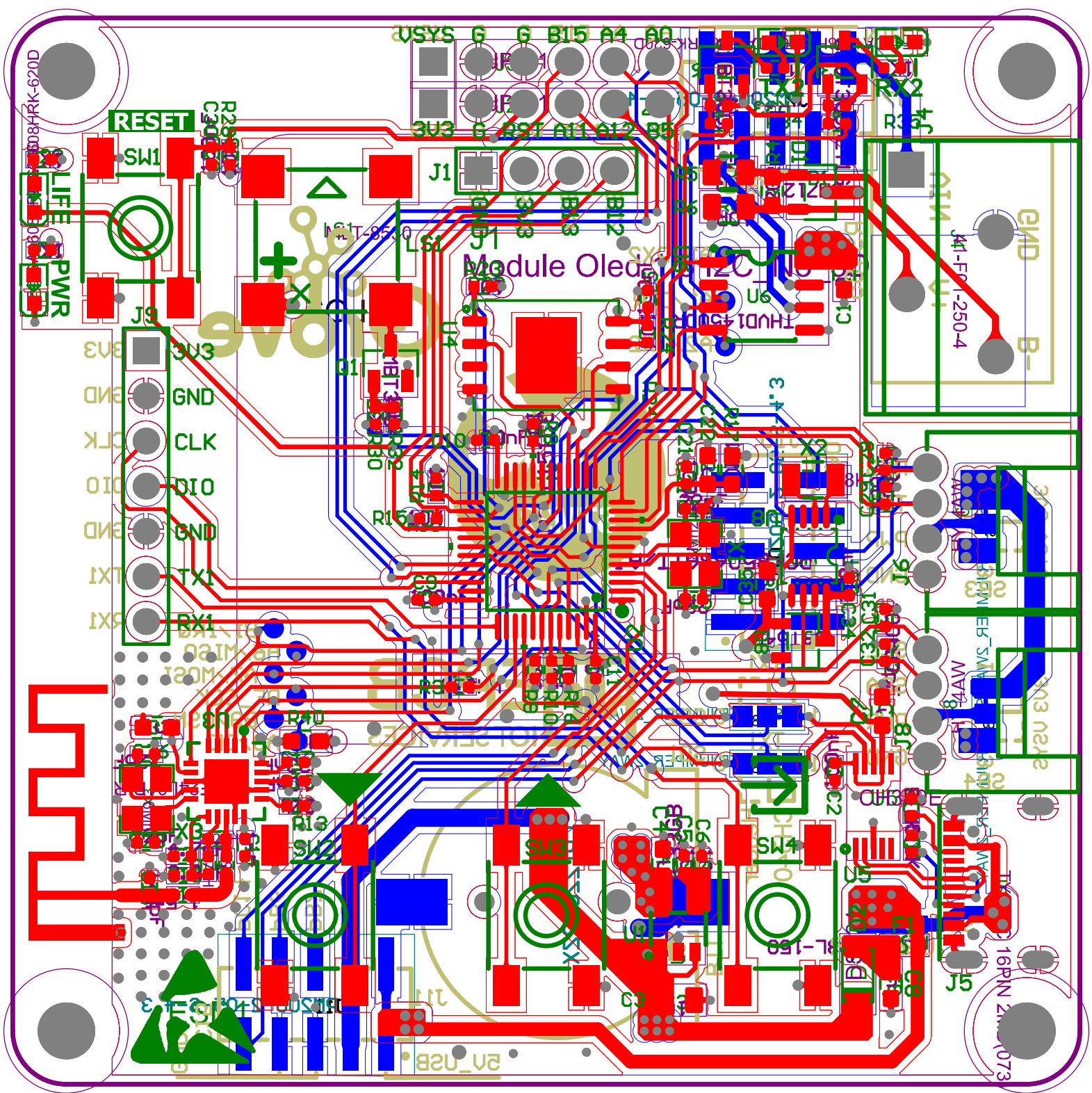


NOR Flash



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AK Base Kit 2.2.PrjPcb

#	Comment	Designator	Footprint	CSO Part Number	Manufacturer Part Number	Manufacturer	Quantity	Description	Assembly Type	Link
1	CR1220-2ZX	BT1	BAT_CR1220_SMD	C969906	CR1220-2ZX	Q&J	1	SMD Battery Holders, C	SMT Assembly	
2	0.1uF	C1	C_0603	C14663	CC0603KRX7R9BB104	YAGEO	1	100NF 50V 0603 ROHS	SMT Assembly	
3	100nF	C2, C9,	C_0402	C41851	0402B104K160NT	FH(Guangdong Fenghua Adv	8	100nF 16V X7R ±10% C	SMT Assembly	
4	10uF	C3	C_0805	C15850	CL21A106KAYNNNE	Samsung Electro-Mechanics	1	10UF 25V 0805 ROHS	SMT Assembly	
5	1uF	C4, C7,	C_0603	C15849	CL10A105KB8NNNC	Samsung Electro-Mechanics	5	1UF 50V 0603 ROHS	SMT Assembly	
6	10nF	C5, C6,	C_0402	C1547	0402CG120J500NT	FH(Guangdong Fenghua Adv	3	10nF 50V X7R ±10% 04	SMT Assembly	
7	33nF	C12	C_0402	C106862	CC0402KRX7R9BB333	YAGEO	1	50V 33nF X7R ±10% 04	SMT Assembly	
8	1nF	C14	C_0402	C1523	0402B102K500NT	FH(Guangdong Fenghua Adv	1	1nF 50V X7R ±10% 04	SMT Assembly	
9	22pF	C15, C1	C_0402	C1555	0402CG220J500NT	FH(Guangdong Fenghua Adv	4	22pF 50V C0G ±5% 04	SMT Assembly	
10	1.5pF	C17	C_0402	C1552	0402CG1R5C500NT	FH(Guangdong Fenghua Adv	1	1.5pF 50V C0G ±0.25pF	SMT Assembly	
11	1.0pF	C18	C_0402	C1550	0402CG1R0C500NT	FH(Guangdong Fenghua Adv	1	1.0pF 50V C0G ±5% 04	SMT Assembly	
12	2.2nF	C19	C_0402	C1531	0402B222K500NT	FH(Guangdong Fenghua Adv	1	2.2nF 50V C0G ±5% 04	SMT Assembly	
13	4.7pF	C20	C_0402	C1569	0402CG4R7C500NT	FH(Guangdong Fenghua Adv	1	4.7pF 50V C0G ±5% 04	SMT Assembly	
14	100pF	C28, C2	C_0402	C1546	0402CG101J500NT	FH(Guangdong Fenghua Adv	4	100pF 50V C0G ±5% 04	SMT Assembly	
15	4.7uF	C35	C_0603	C19666	CL10A475KO8NNNC	Samsung Electro-Mechanics	1	4.7UF 16V 0603 ROHS	SMT Assembly	
16	SM712	D1	SOT-23-3_L2.9-W1.1	C502564	SM712	MDD (Microdiode Electronics	1	TVS DIODE 12V 21V S	SMT Assembly	
17	DSK34	D2	SOD-123_L2.8-W1.8	C41029	DSK34	MDD (Microdiode Electronics	1	DSK34 40V 550mV@3	SMT Assembly	
18	FC-DA1608HRK-620	D4, D5,	LED0603-RD_RED	C84263	FC-DA1608HRK-620D	NATIONSTAR	4	LED Red 0603 Light Em	SMT Assembly	
19	BAT54C	D8	SOT-23-3_L2.9-W1.1	C14308	BAT54CLT1G	ON Semicon	1	SCHOTTKY BARRIER	SMT Assembly	
20	BSMD0603L-150	F1	F0603	C914094	BSMD0603L-150	BHFUSE	1	6V 1.5A 50A 1W -40°C-	SMT Assembly	
21	Module Oled 1.3 I2C	J1	Module Oled 1.3 - No	N/A	N/A	EPCB.vn	1	Module Oled 1.3 I2C	Hand Soldering	
22	41-F01-250-4	J4	CONN-TH_41-F01-2	C173392	41-F01-250-4	HIWA	1	Terminal Block 4 0.138	Hand Soldering	
23	TYPE-C 16PIN 2MD	J5	USB-C-SMD_TYPE	C2765186	TYPE-C16PIN	SHOU HAN	1	5A 1 Surface Mount 16	SMT Assembly	
24	HY-4AW	J6, J8	CONN-TH_4P-P2.0	C146068	HY-4AW	BOOMELE	2	HY-4AW 1x4P 1 2mm 4	Hand Soldering	
25	PM200-2-05-S-4.3	J7, J10,	HDR-SMD_10P-P2.0	C3975169	PM200-2-05-S-4.3	HCTL(华灿天禄)	3	SMD,P=2mm Female H	SMT Assembly	
26	HDR1x7	J9	HDR1X7H		HDR1x7 Horizontal	China	1	Header IDC, 1x8 Pin ho	Hand Soldering	
27	3.9nH	L1	L_0402	C317939	FHW0402UC3N9JST	FH(Guangdong Fenghua Adv	1	3.9nH 840mA ±5% 66m	SMT Assembly	
28	8.2nH	L2	L_0402	C317946	FHW0402UC8N2JST	FH(Guangdong Fenghua Adv	1	8.2nH 680mA ±5% 100	SMT Assembly	
29	2.7nH	L3	L_0402	C317935	FHW0402UC2N7KST	FH(Guangdong Fenghua Adv	1	2.7nH 640mA ±10% 12	SMT Assembly	
30	MLT-8530	LS1	BUZ-SMD_4P-L8.5-V	C94599	MLT-8530	Jiangsu Huaneng Elec	1	BUZZERS MAGNETIC	SMT Assembly	
31	MMBT3904	Q1	SOT-23-3_L2.9-W1.1	C20526	MMBT3904	Changjiang Electronics Tech (	1	MMBT3904 TRANSISTO	SMT Assembly	
32	MMBT3906LT1G	Q2, Q3	SOT-23-3_L2.9-W1.1	C53444	MMBT3906LT1G	ON Semicon	2	MMBT3906 TRANSISTO	SMT Assembly	
33	220HM	R3	R_0402	C25092	0402WGF220JTCE	Uniroyal Elec	1	22Ω 0402 62.5mW Thic	SMT Assembly	
34	120R	R4	R_0603	C22787	0603WAF1200T5E	UNI-ROYAL(Uniroyal Elec)	1	120 OHMS ±1% 1/10W	SMT Assembly	
35	10R	R5, R6	R_0805	C17415	0805W8F100JT5E	Uniroyal Elec	2	10OHMS ±1% 1/8W 08	SMT Assembly	
36	22k	R7, R13	R_0402	C25768	0402WGF2202TCE	Uniroyal Elec	3	22k 0402 62.5mW Thic	SMT Assembly	
37	10K	R8, R9,	R_0402	C25744	0402WGF1002TCE	Uniroyal Elec	10	10kΩ 0402 62.5mW Th	SMT Assembly	
38	5.1k	R11, R1	R_0402	C25941	0402WGF0512TCE	Uniroyal Elec	2	5.1KOHMS ±5% 1/16W	SMT Assembly	
39	00HM	R17, R4	R_0603	C21189	0603WAF0000T5E	Uniroyal Elec	2	00HMS ±1% 1/10W 06	SMT Assembly	
40	1k	R29, R3	R_0402	C11702	0402WGF1001TCE	Uniroyal Elec	7	1kΩ 0402 62.5mW Thic	SMT Assembly	
41	TS-1102S-B-G-B	SW1, S	SW-SMD_4P-L6.0-V	C381064	TS-1102S-C-F-B	XKB Connectivity	4	Bottom SMD TACTILE S	SMT Assembly	
42	RT9193-33GB	U1	SOT-23-5_L3.0-W1.1	C15651	RT9193-33GB	Richtek Tech	1	5.5V 300mV @ 300mA	SMT Assembly	
43	STM32L151C8T6	U2	LQFP-48_L7.0-W7.0	C9861	STM32L151C8T6	STMicroelectronics	1	STM32 ARM-based 32-	SMT Assembly	
44	NRF24L01P-R	U3	QFN-20_L4.0-W4.0	C8791	NRF24L01P-R	NORDIC	1	NRF24L01P-R SPI QFN	SMT Assembly	
45	W25Q256JVEIQTR	U4	WSO8-8_L8.0-W6.1	C97522	W25Q256JVEIQTR	Winbond Elec	1	W25Q256 WSON-8 8x	SMT Assembly	
46	CH340E	U5	MSOP-10_L3.0-W3.0-P0.50-LS5.0-BL			WXH(Jiangsu Qin Heng)	1	Transceiver USB 2.0 2	SMT Assembly	
47	THVD1450DR	U6	SOIC-8_3.9x4.9x1.2	C2671361	THVD1450DR	Texas Instruments	1	THVD1450DR transcei	SMT Assembly	
48	PCF85063ATT/AJ	U8	TSSOP-8_L3.0-W3.1	C165608	PCF85063ATT/AJ	NXP	1	TSSOP-8-3.0mm Real	SMT Assembly	
49	8Mhz	X1	SMD-3225_4P	C133366	1C208000CE0Q	KDS Daishinku	1	8Mhz 3225 SMD CRYST	SMT Assembly	
50	32.768K	X2	SMD-3215_2P	C32346	Q13FC1350000400	Seiko Epson	1	32.768K SMD CRYSTA	SMT Assembly	
51	16Mhz	X3	SMD-3225_4P	C13738	X322516MLB4SI	Yangxing Tech	1	16MHz SMD Crystal Re	SMT Assembly	